

Avionics Design Architecture for Low-Cost CubeSat Missions and Lessons Learned from R5-S2 and R5-S4

Kathryn Knesek
NASA Johnson Space Center
2101 E NASA Pkwy
Houston, TX 77058
Kathryn.Knesek@nasa.gov

Morgan Alexander
NASA Johnson Space Center
2101 E NASA Pkwy
Houston, TX 77058
Morgan.Alexander@nasa.gov

Jack Wisbiski
NASA Johnson Space Center
2101 E NASA Pkwy
Houston, TX 77058
Jack.s.Wisbiski@nasa.gov

Abstract— The Realizing Rapid, Reduced-cost high-Risk Research (R5) project is primarily intended to develop and operate a capability to host low technology readiness level (TRL) payloads and demonstrations much faster and cheaper than the current state of the art. The primary stakeholder for this project is the Small Spacecraft Technology program within the Space Technology Mission Directorate at NASA with the project based out of Johnson Space Center. Unlike typical NASA satellites, R5's spacecrafts utilize many commercial off-the-shelf (COTS) components while designing custom hardware only when necessary. This paper explores the avionics subsystem design philosophy, the resulting design, and the lessons learned throughout the design, build, test, and operations of R5 Spacecraft 2 and R5 Spacecraft 4 (R5-S2 and R5-S4).

The avionics subsystem has divided its design into three main functional groups: power, propulsion control, and payload interfaces. The power group contains the solar panels, batteries, and a battery management system. The propulsion control group contains the electronics needed to control thrusters, monitor propulsion systems, and operate reaction wheels. The payload interfaces group contains a variety of standard COTS interfaces to facilitate the integration of a broad range of low TRL prospective payloads (not requiring payloads to develop to a singular data interface). The avionics subsystem design supports late changes to mission goals and spacecraft configuration, reducing the need for redesign and testing time.

Typical CubeSat architectures use the PC/104 specification, a modular framework in which avionics PCBs are stacked. As the R5 project utilizes many COTS components and interfaces with a wide variety of payloads, the PC/104 specification does not meet the needs of the project. Instead, the R5 avionics subsystem has developed an alternative framework that allows for flexible mechanical and electrical integration with this variety of components.

Throughout the design, build, and test process of R5-S2 and R5-S4, the team learned many lessons that were then applied to future designs (spacecrafts 3 and 5 and beyond). This paper will highlight the lessons learned and how the subsystem design has evolved to reduce risk for future missions and enhance capability.

TABLE OF CONTENTS

1. INTRODUCTION TO THE R5 PROJECT	1
2. R5-S2 AND R5-S4 MISSION OVERVIEW	2
3. R5-S2 AND R5-S4 AVIONICS DESIGN AND DEVELOPMENT	2

4. R5-S2 AND R5-S4 AVIONICS OVERVIEW.....	5
5. R5-S2 AND R5-S4 PERFORMANCE IN ORBIT	7
6. AVIONICS LESSONS LEARNED.....	8
7. AVIONICS CHANGES AND FORWARD WORK.....	10
8. SUMMARY	13
REFERENCES	14
BIOGRAPHY	15

1. INTRODUCTION TO THE R5 PROJECT

The Realizing Rapid, Reduced-cost high-Risk Research (R5) project is primarily intended to integrate, launch, and then operate payloads in low Earth orbit (LEO). While there are several existing capabilities that can provide these services, the sheer volume of payloads of interest and the desire to have payloads on-orbit less than a year after selection drive a need for a more rapid and lower cost capability than the existing state of the art. The overall goals of cost and schedule reduction apply to the payloads as well, adding requirements onto R5 to ensure it can easily support technology readiness level (TRL) 4 payloads without requiring them to perform extensive environmental qualification or interface development. R5 has many other objectives including advancing in-space-inspection-relevant technologies, reevaluating many standard processes and protocols surrounding spaceflight hardware development, accelerating assessment of modern commercial-off-the-shelf (COTS) components, sharing cost and schedule saving opportunities and methods with the broader space community, and developing early career engineers. R5's primary stakeholder is the Small Spacecraft Technology (SST) program within the Space Technology Mission Directorate (STMD) at NASA and the project itself is based out of Johnson Space Center (JSC).

To minimize cost and schedule, the R5 project uses a minimum viable product approach to scope definition and uses an iterative and incremental development paradigm. Each spacecraft contains increased capability and addresses lessons learned from previous experience. In order to maximize the value of each mission, secondary payloads and flight objectives are added as able after the identification of a primary payload and objectives. The spacecraft architecture, already designed to be inherently flexible to accommodate a

broad range of payloads while minimizing the non-recurring engineering (NRE) needed to integrate them, simplifies the addition of these secondary payloads and objectives.

R5 started in 2021 as an internal research and development project at JSC. The project was able to build from the lessons learned in JSC's prior Seeker CubeSat project. In late 2021, R5 Spacecraft 1 (R5-S1) was assembled, tested, and delivered for launch onboard Astra's LV0008. R5-S1 was made entirely from COTS parts with a custom aluminum structure. LV0008 launched in early 2022 with R5-S1 onboard but failed to reach orbit. R5-S1 was not heard from during its very brief flight, but many lessons were still learned during development, assembly, integration, and testing (AIT) of the vehicle. During the development of R5 Spacecraft 2 (R5-S2) and R5 Spacecraft 4 (R5-S4), STMD SST became the primary stakeholder for the project.

2. R5-S2 AND R5-S4 MISSION OVERVIEW

R5-S4 was intended to build on the lessons of R5-S1 while also adding a prototype propulsion subsystem. With R5-S1 failing to reach orbit, R5-S2 carried many of the same objectives of gathering data on the results of the ultra-lean process that R5 was pioneering. Additional details of the applied process and its results are detailed in sections below. After some schedule and manifest changes, R5-S2 and R5-S4 were both manifested on the NASA Educational Launch of Nanosatellites (ELaNa) 43 mission. Both were given nearly identical missions, intending to provide additional data over a single spacecraft.

The primary objective and minimum success criteria for these initial R5 missions was for each spacecraft to power on, autonomously start operations, and transmit telemetry to the ground. The secondary objectives included:

- Commanding the spacecraft from the ground
- Demonstrating the cold gas propulsion system
- Automatically shutting down, recharging batteries from solar panels, rebooting, and resuming the mission from the previous operational period
- Validating environment assumptions
- Demonstrating larger messages over Iridium's short burst data (SBD) link
- Imaging the launch vehicle upper stage upon deployment and downlink imagery
- Exercising the guidance, navigation, and control (GNC) system that included star tracker, global positioning system (GPS), and inertial measurement unit (IMU)

The spacecrafts also hosted payloads with both carrying fiducial markers to assist with proximity operations and R5-S4 hosting an Extremely Low Resource Optical Identifier (ELROI) from Los Alamos National Laboratory (LANL).

Several schedule changes enabled a complete rebuild of the spacecrafts between October 2023 and April 2024 to

incorporate improvements from lessons learned during the initial AIT efforts. R5-S2 and R5-S4 were both 6U ("Units" of 10 cm³) spacecraft and can be seen in Figure 1.

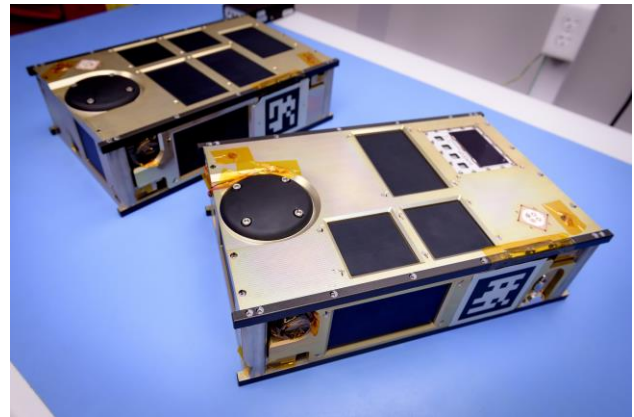


Figure 1: R5-S2 (left) and R5-S4 (right)

Planned scope enhancements for R5-S2 (and beyond) and the lessons learned from R5-S1 demonstrated that custom avionics components were required due to several factors, including the limited power output from COTS components. In order to develop these and manage the growing avionics subsystem, a team of early career engineers from the JSC Avionics Systems Division were brought in. Around the same time, R5-S2 and R5-S4 were manifested on the same launch.

3. R5-S2 AND R5-S4 AVIONICS DESIGN AND DEVELOPMENT

R5-S2 and R5-S4 were designed to share as many common components as possible to minimize cost and schedule. The spacecrafts incorporated the same COTS cameras, modem, GPS, and IMU. The avionics team was brought on to the project to provide custom designs for needs that could not be met by COTS assemblies within the resource limitations of R5. At the beginning of the first development cycle after R5-S1, the scope of custom assemblies was limited to the power system, but before the delivery of the spacecrafts had evolved to include a propulsion system controller, battery pack, and wire harnessing.

Requirements and Interfaces

In typical NASA spaceflight projects, requirements are developed early in the project lifecycle and are quite extensive. The high level of detail can substantially extend the schedule needed to complete requirements definition. Even with this level of detail, it is very hard to write a fully comprehensive set of requirements early in the project lifecycle due to the complexity of space missions, the space environment, and inevitable changes in project direction over time. This can lead to requirements that are no longer relevant or are impossible to verify, adding cost and schedule to the project in the verification phase as well. R5 is

exploring opportunities to reduce schedule by limiting requirement development and instead focusing on instilling all personnel with a deep understanding of each R5 mission and its interfaces. This is enabled by the limited scope of each R5 mission. Even with the focus on reducing requirements and emphasizing the mission, there are still some that must be addressed, including those in the CubeSat Design Specification and any associated launch vehicle requirements [1]. For the avionics subsystem, these are generally associated with hazards, power systems, electrical inhibits, and environmental testing.

Typical NASA processes for subsystem development include the development of interface control documents to capture the definition of the interfaces between subsystems. These documents are often disconnected from the methods subsystems natively use for tracking the evolution of their systems during development. As such, developing and maintaining interface control documents is typically not able to be done directly in-line with subsystem development and takes subsystem resources away from development work on the critical path. R5 is exploring ways to reduce schedule by allowing subsystems to maintain their own documentation as they see fit without dictating a specific standardized method or format for interface definition. All R5 subsystems maintained a source of truth in a format that made sense for their workflow. The avionics subsystem source of truth was a chart set detailing information relevant to others: connector pinouts, mechanical information (e.g., mounting fastener provisions, dimensions, backshell dimensions and minimum bend radius for cables), computer bus and pin allocations, and power availability. Other subsystems similarly defined their interfaces in documents close to their own workflows. Most subsystem leads were co-located in the lab workspace, making it easier to work through interfaces in person to reduce misunderstandings.

Environmental Considerations

Designing for spaceflight typically means considering environmental conditions such as mechanical shock and vibration (the launch environment), large temperature ranges, vacuum, and radiation (the space environment). R5's goal of pioneering a leaner process, high risk acceptance, ability to leverage deep institutional knowledge at JSC around component selection for space, and recent environmental test results of COTS components all enabled the project to descope all of the environmental testing with the exception of random vibration (as it is required by launch providers) [2][3][4][5].

Assembly Selection

For all avionics subsystem functions, COTS assemblies were first considered for use. A market survey was performed on available industrial, commercial, medical, and tactical-grade COTS assemblies for each major avionics functional area, focused on the functional capability of the assemblies, their cost, and lead time. Where suitable COTS assemblies were

identified, they were selected and acquired. If there was no previous information on their performance in various space environments (or no performance analogies existed with similar models, manufacturers, or construction), some limited environmental testing was performed to verify their ability to briefly survive vacuum and thermal ranges beyond their rated ranges. For avionics functions that could not be met with COTS assemblies, the project had to create custom designs to stay within its resource limitations.

Custom Avionics Development Process

Custom assemblies were developed with an iterative process, driving towards a functional yet minimal viable product. Development started with the identification of key components that were required to enable functionality (e.g., a battery management integrated circuit). With no other hard environmental requirements, the avionics subsystem aimed to increase reliability of custom designs while maintaining cost and schedule constraints. Automotive-grade parts (passives and integrated circuits) were used wherever possible, or parts with an extended temperature range matching automotive grade 3 (-40 °C to +85 °C) or better. Non-metallic materials were tracked by the structures subsystem, and low outgassing components were used for all connectors and board components. Wire jackets and shrink tubing were all polytetrafluoroethylene (PTFE), which is typically used by NASA and has low outgassing properties. Cable ties were ethylene-tetrafluoroethylene (ETFE), which has similar material properties. All cabling conformed to military specification M22759. Next, subassemblies were prototyped using evaluation boards and/or breadboards. Prototype printed circuit boards (PCBs) were then designed using Altium Designer, fabricated at a commercial fabricator, and then assembled in-house using a DMM Novastar LE-40V pick and place machine.

Specifically for the battery pack, a 1S pack design was preferred as it greatly simplified charge safety compared to packs with multiple cells in serial. Underwriter's Laboratories-rated 18650 Lithium-Ion cells were baselined due to the familiarity of launch vehicle providers and launch ranges with them, which streamlined safety discussions. Tab welding was used for cell connectivity due to its robustness to the vibration environment. The total pack energy storage was designed to fit within Transportation Safety Administration guidelines for airline carry-on to enable additional flexibility in spacecraft transportation.

PCBs were fabricated by quick-turnaround commercial fabricators such as Sunstone Circuits. Wherever possible, designs were standardized to meet Sunstone Circuits' rapid quote/turn requirements, including maintaining minimum drilled hole sizes, trace widths, and using standard layer stack-ups. The board material was FR-4, with tin lead or electroless nickel immersion gold (ENIG) finish. According to NASA's Materials and Process Selection, Control, and Implementation Plan for JSC Flight Hardware, JSC 27301F, the fabrication process for PCBs shall meet the PRC-7005, or

IPC-6011, and IPC-6012 [6]. The PCBs should be fabricated to IPC class 3 with space addendum IPC-6012-ES, but R5 accepted the risk of using Sunstone Circuits' standard IPC-A-600 Class 2 fabrication to significantly reduce schedule and cost [7].

Checkout procedures were written for each board subassembly to check for manufacturing defects and functionality. After passing checkouts, completed boards were put through thermal cycling to verify workmanship. A TestEquity TE-107 benchtop thermal chamber was used to perform cycling to temperatures based on a preliminary thermal analysis of the orbital environment. The cycle profile was -40 °C to +60 °C with a maximum ramp rate of 4 °C per minute, an hour dwell at the hot and cold temperature targets, and three complete cycles. This thermal cycling was a low time and effort impact to the team, with each cycle set taking roughly half a day to complete. This test reliably caught unexpected solder joint fatigue, allowing rework to prevent failure in random vibration testing and/or in orbit. Functional checkouts were repeated after thermal cycling. After passing these tests, the assemblies were moving into integrated testing. This was done with a representative system flat satellite (FlatSat) shown in Figure 2, consisting of the contents of a full spacecraft mounted on an optical breadboard instead of a tightly integrated structure. System integration as early as possible allowed more time for full-system functional checkouts and identification of design issues. Discovered issues were addressed in a redesign of the custom assemblies.

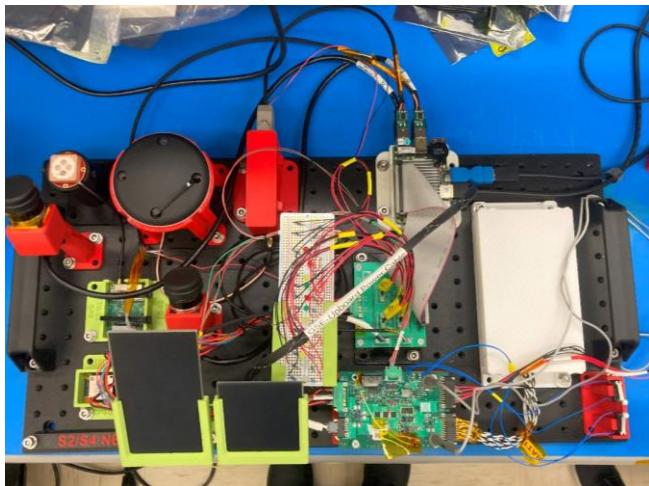


Figure 2: FlatSat used for testing both R5-S2 and R5-S4

After successful checkouts, PCBs were conformally coated with Dow's DOWSIL 3140 room-temperature vulcanizing silicone (RTV) and large components and connectors were staked with DOWSIL 3145 RTV adhesive/sealant to provide mechanical relief which can be seen inside of R5-S2 in Figure 3. NASA projects typically follow NASA-STD-8739.6 and NASA-STD-8739.1 for workmanship when conformal coating. R5 generally followed these standards, but replaced

the typical formal training with project-level training and did not require vision screening.

System Assembly

System assembly was performed by the structures subsystem, but is described here as it provides additional insight into how avionics components were handled prior to flight. For system assembly, all subsystems delivered completed assemblies to the structures subsystem. These assemblies included integrated cameras and lenses, completed PCBs, pressure manifolds with valves and sensors installed, wire harnesses, and more. The assembly was done in an office-like shirt-sleeve environment with basic foreign object debris (FOD) controls (not a cleanroom). Assembly was performed on grounded surfaces, but those performing assembly did not use wrist straps to ground themselves to the work surface. The assembly area itself was also not verified for electrostatic discharge (ESD) compliance (e.g., flooring, chair grounding, etc.). After assembly, the spacecrafts were handled through a variety of test environments without ESD protection.

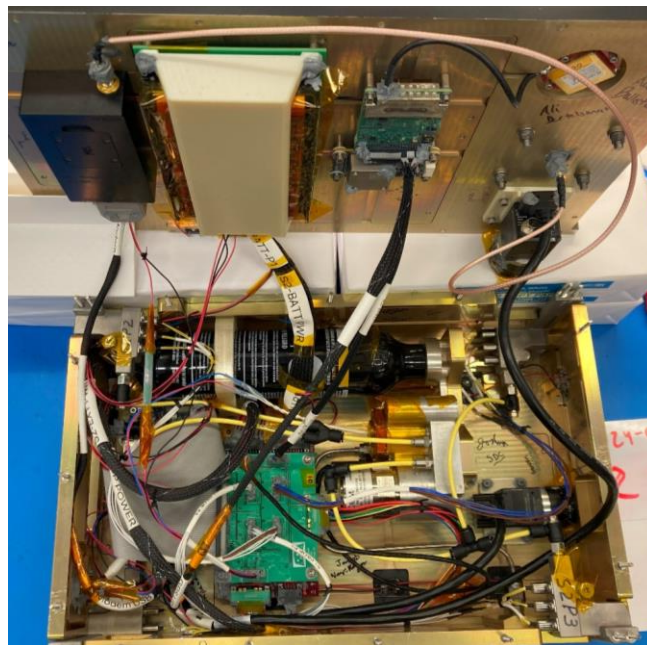


Figure 3: R5-S2 spacecraft interior

System Verification

After the verification of the full-system functionality and any final component modifications (e.g., conformal coating), “day-in-the-life” (DITL) testing was performed on the FlatSat. This involved the hardware and software operating through all mission phases as best as they could be approximated on the ground, including operating with a “live-sky.” This demonstrated the system’s performance through the mission phases as it automatically transitioned through them and performed end-to-end communication via the Iridium network.

Before the two spacecrafts were delivered, the only spacecraft-level environmental testing that occurred was random vibration testing. Complete spacecraft functional checkouts were performed before and after random vibration testing to ensure that no components experienced a change in functionality. The various stakeholders did not require thermal vacuum, mechanical shock, or electromagnetic interference (EMI) testing. As part of R5's lean process goals, individual subsystems did further tests only as they deemed necessary.

After performing all environmental and functional testing, the integrated spacecrafts were run through live-sky DITL testing, as pictured in Figure 4. In verifying the end-to-end function of the system, this also verified the system didn't create any EMI problems for itself.



Figure 4: DITL live-sky testing of R5-S2 (upper right) and R5-S4 (center right)

4. R5-S2 AND R5-S4 AVIONICS OVERVIEW

The R5-S2 and R5-S4 avionics subsystems were divided into the following functional groups: flight computer, power, propulsion control, and peripherals and payloads.

Flight Computer

The flight computer was the UP Board UP-CHT01-A20-0464-A11. This COTS single board computer has an Intel Atom x5-Z8350 system on a chip with an Intel Gen8 HD400 internal GPU, 4GB of DDR3 memory, and 64GB of eMMC storage. The interfaces available for use include four USB 2.0 ports, one HDMI port, one USB 3.0 port, one ethernet port, UART/SPI/I2C serial interfaces, and up to 28 general-purpose input/output (GPIO) pins.

Power

The power assembly included the power distribution unit (PDU), solar panels, batteries, battery recharge, and power safety features.

Power Distribution Unit—The PDU was a custom PCB assembly responsible for charging and monitoring the batteries, protection against power faults, and providing 5 V, 12 V, and 3.3 V power rails for the rest of the avionics. R5 baselines the support of high performance and high power consumption flight computers and payloads as opposed to comparable spacecrafts. This PDU needed to support a current draw of 10 A. A custom design was created as the commercial options could not meet the power requirements, were too expensive, had lead times that were too long, or didn't work with the size constraint.

The PDU uses a Texas Instruments (TI) BQ27Z746 battery gauge to act as a smart fuse and report battery status telemetry. The temperature controls and low power modes were turned off for this chip. The PDU battery charge circuit uses a TI BQ25798 and is configured to take input from an external power supply (ground use only) or the solar panels. The solar panel control circuitry includes an Omron G6BU-1114P-US-DC12 latching relay on the PDU which is commanded to disconnect the panels from the charge circuitry at the end of mission to disable charging capability, meeting passivation requirements. The output voltage rails are created by TI TPS61288s in parallel and turned on or off by an Analog Devices LTC4418 in conjunction with a Microchip Technology MIC2778 that is monitoring the battery voltage. The TI INA3221 current monitor reports information about spacecraft power usage to the ground. The functional block diagram of the PDU is shown in Figure 5.

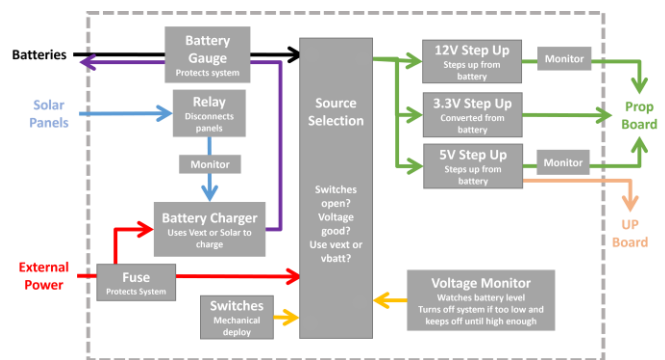


Figure 5: PDU system diagram for R5-S2 and R5-S4

Shown in Figure 6, the PDU used connectors from the Harwin M80 family (black connectors) for power and connectors from the Harwin G125 series (green connectors) for data.

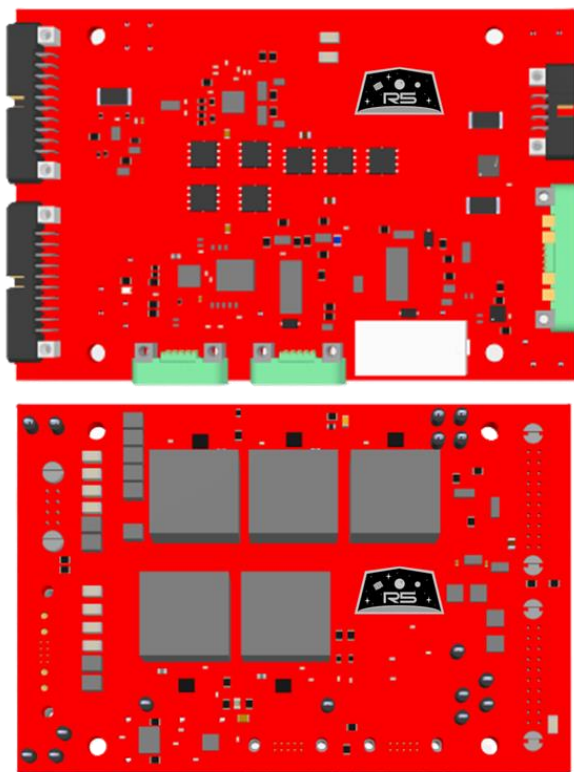


Figure 6: Top and bottom views of the PDU PCB for R5-S2 and R5-S4

Solar Panels—Voltaic Systems P123 0.6 W, 6 V (66x66 mm) and P124 1.2 W, 6 V (113x66 mm) ETFE coated solar panels were body-mounted across the spacecraft wherever possible. These panels reported cell efficiency at 22% and 0.5-0.6 mW/cm² output [8][9]. The small form factor provided additional flexibility to the structures subsystem for their placement.

Batteries—R5-S2 and R5-S4 each contained a 1S6P (1-series, 6-parallel cells) pack consisting of Samsung 30Q 18650 3,000 mAh Li-ion batteries. These provide 15 A continuous discharge rating and were familiar to JSC’s Propulsion and Power Division. The final configuration of the battery (Figure 7) was a tab-welded design (custom manufactured by KULR Technology Group) that fit within a custom PCB and 3D printed housing.



Figure 7: Battery pack design for R5-S2 and R5-S4

Propulsion Control

The CubeSats contained a cold-gas propulsion system that required control of twelve Parker C7 miniature cartridge solenoid thruster valves, one Lee Company high-pressure isolation solenoid valve, and one TE Connectivity EB100 high accuracy miniature pressure transducer. The propulsion system elements were managed by the R5 propulsion subsystem, but control and electrical interfaces were managed by the avionics subsystem.

A propulsion controller custom PCB (referred to as the “prop board”), Figure 8, was designed to interface the flight computer with the propulsion subsystem’s electromechanical components. This board took in the flight computer’s 40-pin header containing GPIO and serial interfaces and controlled fourteen Monolithic Power Systems’ automotive-grade MPQ6610 half-bridge drivers for the high-pressure isolation and thruster valves.

In addition to the propulsion system, the prop board used the half-bridge switches to provide switched 5 V power to the higher-power modem to ensure it could be completely powered off to meet mission passivation requirements.

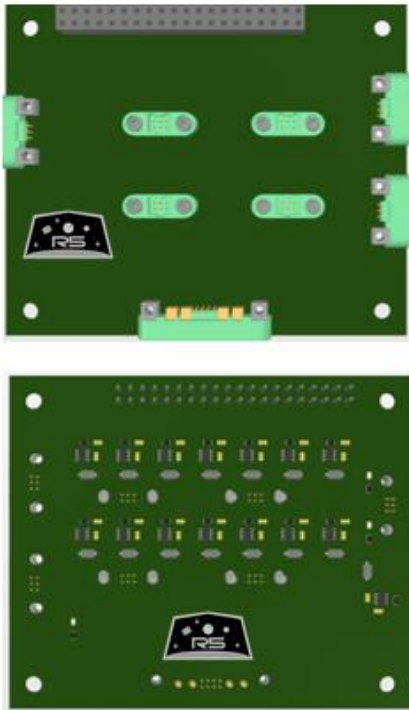


Figure 8: Top and bottom views of the prop board PCB for both R5-S2 and R5-S4

Cameras and Peripherals

Other peripherals that the avionics subsystem powered and interfaced the computer to included:

- 1x Epson M-G370PDS0 IMU
- 1x Ximea MQ013CG-E2 camera
- 1x Ximea MC023MG-SY-UB camera
- 1x JAVAD TR-2S GPS receiver
- 1x NAL A3LA-RS Iridium Modem

The cameras, modem, and GPS were connected to the flight computer by USB. The IMU used a UART serial interface routed through the prop board.

5. R5-S2 AND R5-S4 PERFORMANCE IN ORBIT

R5-S2 and R5-S4 were delivered for deployer integration on April 24, 2024, and launched on July 3, 2024 on Firefly's Alpha FLTA005 as part of the Venture-Class Launch Services (VCLS) Demo 2-FB / ELaN 43 mission. Upon deployment from the rocket, both spacecrafts booted up from battery power and started sending data packets to the ground over the Iridium link, completing the primary objective.

In their first operational period, both spacecrafts responded to commands from the ground and completed a propulsion system demonstration, successfully demonstrating the function of all parts of the propulsion system. Upon running the battery down to the PDU's shutdown threshold, the

spacecrafts powered off and the batteries began to recharge. Hours later, R5-S2 and R5-S4 automatically powered back on and resumed the mission. The spacecrafts have continued to operate in this fashion and additional demonstrations have been completed during their time in orbit, including evaluating the limits of the Iridium communication system (repeatedly successfully sending 1 kB SBD messages, which is approximately three times greater than previous missions), downlinking of imagery from the cameras as expected, and enabling LANL to successfully observe the ELROI payload.

The direct avionics subsystem telemetry data downlinked from R5-S2 and R5-S4 has been less than expected. Limited fault analysis suggests this is due to a software error that inconsistently causes the flight software application that collects this telemetry to crash. Despite this, sufficient data has been gathered to draw conclusions about the environment and the system's performance.

The spacecrafts continue to gather temperature data of the PDU, flight computer, and battery pack, validating environment assumptions. As seen in Figure 9, R5-S2's battery pack temperature was measured to be between 19 °C and 54 °C, while R5-S4's was measured to be between 22 °C and 47 °C, both of which are within the Samsung 30Q's rated discharge temperature range of -20 °C to 75 °C.

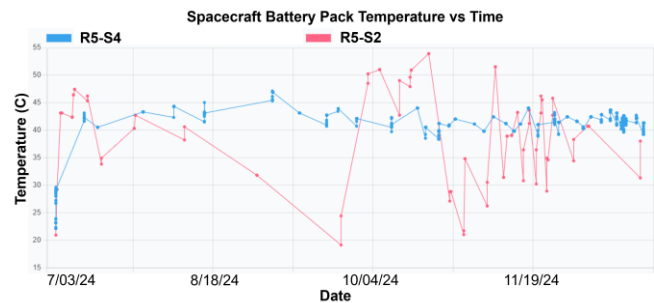


Figure 9: Temperature of R5-S2 and R5-S4 battery packs in orbit

Over their first 6 months in space, R5-S2 has successfully booted 33 times and R5-S4 has successfully booted 49 times. The spacecrafts' recharge times have varied considerably. R5 suspects this is related to orbit lighting conditions and spacecraft attitude as opposed to power system performance or degradation, though more analysis is needed. In Figure 10, there are several instances where the spacecraft boot counter increments by more than one. This signifies that there was an operational cycle where the spacecraft successfully booted and started its flight software (FSW) but did not successfully transmit data to the Iridium network (and therefore to the ground as well).

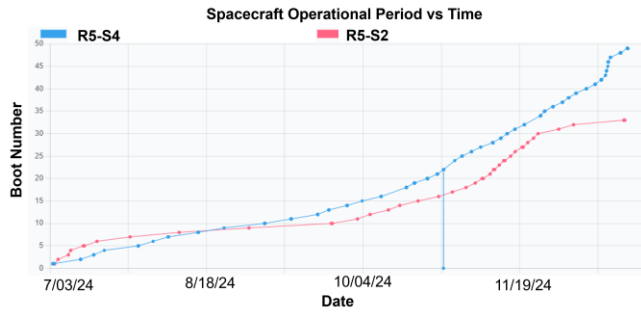


Figure 10: Number of boot cycles for R5-S2 and R5-S4 in orbit

During the spacecraft's initial operational period, where their batteries were at their highest state of charge across the mission, R5-S4 operated for approximately 5.4 hours and R5-S2 operated for approximately 5.9 hours. This data is the time of the last telemetry message from the spacecrafts during the first operational period and not the true duration of the period as it isn't recorded. This is the best data available to assess a minimum operational time for each operational period, though it must be evaluated with the caveat that many other factors (like Iridium network geometry) can impact successful data transmission beyond spacecraft power. During subsequent operational periods, R5-S2's last successful telemetry transmission is approximately 2.2 hours on average into its operational periods while R5-S4's last successful telemetry transmission is approximately 2.8 hours on average into its operational periods. The time from boot to last telemetry message sent is shown in Figure 11.

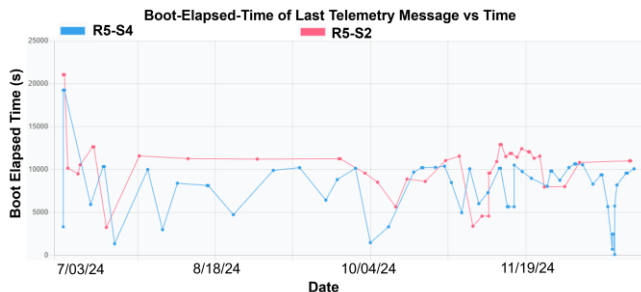


Figure 11: Boot-elapsed-time of last telemetry message from R5-S2 and R5-S4 in orbit

As of December 20th, 2024, both spacecrafts had last successfully completed an operational cycle on December 20, 2024. While the spacecraft remain operational, the R5 team will continue to gather data on the performance of the components, measure the environment, evaluate the consistency of the Iridium network connection, and downlink additional imagery. This will continue until the spacecrafts either stop responding or the radio frequency licenses for the spacecrafts expire, at which point they will be commanded to passivate (drain batteries, stop transmitting, disconnect solar panels from battery, and vent the propellant tank).

6. AVIONICS LESSONS LEARNED

Many lessons across all subsystems were learned during the design and AIT of R5-S2 and R5-S4. Some were able to be incorporated into the spacecrafts themselves while the rest will be applied to future spacecrafts.

Custom Design Lessons Learned

R5 relied on the Saturn PCB toolkit to make decisions about PCB design parameters like trace width, via sizing, etc. However, this tool was not intended for the space environment and calculations that included thermal aspects are not accurate in vacuum. Following existing NASA standards for derating proved important in PCB design, component selection, and wire gauge selection in harnesses [10].

During the fabrication of custom PCBs and assemblies, fabrication failures were relatively common. Immediate inspection of all arriving hardware became a necessity to catch failures early in AIT. One shipment of PDU PCBs had three boards with connector through-holes filled with tin and were unusable. A different order of custom boards had an internal short between power and ground planes due to errors in fabrication. In order to compensate for inconsistencies in PCB fabrication and other errors in assembly, R5 adopted a rule to always order at least twice as many PCBs as were expected to be used. R5 now typically orders 10 PCBs when only four are needed.

In-house board assembly was substantially slowed by using parts not packaged on reels. R5 now baselines procuring parts on reels. Although the R5 in-house assembly process has the capacity for using leadless parts, it proved difficult, especially when using die-sized packages, to ensure proper soldering of the parts (and for any rework). All future R5 custom designs will carefully consider the packaging type of each component in the event those components need to be removed or replaced.

Power System Lessons Learned

Grounding Scheme—The avionics design called for a single-point grounding scheme based on the NASA handbook for grounding in unmanned spacecraft [11]. Accomplishing this grounding scheme was challenging due to the number of COTS parts in the system with mounting holes that are tied to ground. This required insulating the COTS parts from the spacecraft structure (often accomplished with an Ultem 3D printed bracket). This caused the COTS components to take up more volume in the spacecrafts and in some cases required modifying the COTS hardware, undermining the plug-and-play nature that was originally desired.

Battery Connection—Standard 18650 package batteries have numerous COTS options for mounting. Due to the price and abundance of clip-in/spring mounts and their ubiquitous terrestrial use, using a design with these type of battery

mounts was a design choice originally made with the idea that the batteries could be secured by an external housing design and remain easily serviceable. After completing vibration testing, a substantial decrease in pack energy capacity was sometimes observed. After a vibration test of R5-S4, the spacecraft's operational time during functional checkout was less than 8% of what it was pre-test. It was determined that at least half of the batteries in the pack had lost electrical contact with the clips despite the retention strap and external pack housing. R5 now baselines tab welded battery packs to ensure electrical connectivity between all the cells.

Fusing—The PDU had a fuse and other protections for the batteries, but the pack itself didn't have any protection upstream of its harness (on the pack itself). During assembly, a wire harness was pinched by the structure, puncturing the insulation and causing a ground short of the battery pack to the spacecraft structure. R5 now baselines battery packs to include a fuse on the low side before the wire harness.

Solar Panels—The solar panels the R5 system uses are 6 V panels of varying size and corresponding power output. Though their performance was adequate, the 6 V output caused some design limitations, and 12 V panel options will be investigated in the future. The original solar panels were Voltaic Systems Urethane panels. During AIT, supply chain issues caused challenges procuring these panels, forcing the project to pivot to their ETFE panels. R5's flexible architecture enabled the switch with minimal challenges and the ETFE panels' smaller form factor enabled the structures subsystem additional flexibility in panel placement.

The solar panels, located on all external walls, were soldered together into a single harness. This caused significant challenges during assembly and disassembly, making it impossible to separate the walls of the spacecraft. For future designs, each spacecraft body face will have all of its solar panels (and only its solar panels) connected and connectorized.

Additionally, if external payloads that take up wall surface area are added to the spacecrafts, this reduces the usable area for solar panels, increasing battery recharge time. R5 is evaluating deployable solar arrays, allowing for more payload and solar panel real estate, as well as incorporating maximum power point tracking to improve recharge efficiency.

PDU Design—The PDU handles high current (>10 A) and was designed to be robust. However, standard PCB thicknesses must be used to work with many through-hole components and connectors, thus limiting copper weight and layer counts. For example, 2 oz copper was desired for fabrication, but the battery gauge footprint was a die-size ball grid array (DSBGA) and too small to accommodate the copper greater than 1 oz. This single part significantly increased the difficulty of fabrication and cost.

The overall complexity of the custom PDU was a challenge during AIT. The battery charger and management chips were being tested and reconfigured through the end of the allotted testing period (approximately 72 hours before spacecraft delivery). This served as a reminder to keep the complexity of custom avionics as low as possible to minimize time spent configuring and troubleshooting components.

Propulsion System Lessons Learned

The first edition of the prop board for R5-S2 and R5-S4 used Monolithic Power Systems MPQ6610 half-bridge gate drivers to power the solenoid valves used by the R5 cold-gas thruster system. These gate drive circuits performed high-side switching, disconnecting the positive power line for safety when turning the thrusters or high pressure isolation valve off. This design was very inefficient because it would allow for a greater power draw than what was needed to hold the valves open, and as a result, the thrusters and isolation valve would heat up quickly. During one instance of testing, the isolation valve was left open for a long period of time, resulting in the valve failing open. It was decided to pursue more efficient driving techniques on future spacecrafts, such as pulse width modulation (PWM) or hit-and-hold.

Wire Harness Lessons Learned

R5 utilizes Altium Vault, a design data management system, for PCB design management and version control. Initially the avionics subsystem used Altium's wire harness project type, however the Altium Vault version available did not support this type of project. This affected the drawing version management of the wire harnesses, which lead to out-of-date drawings being used and impacting integrated testing. R5 is now using a different project type to maintain version control on wire harness drawings and maintain compatibility with the version of Altium Vault.

During custom wire harness building and testing, R5 experienced a high rate of wire harness failures and shorts. R5 contacted connector manufacturers and local subject matter experts to completely overhaul the R5 wire harness fabrication process. All harnesses and connectors are labeled, all wire is insulated with PTFE, certain crimps are done with the wire jacket stripped back from the crimp point, all connectors are carefully inspected for FOD and shorted before being backpotted with DOWSIL 3145 RTV, all wire harnesses are verified, and finally all are serialized and tracked.

Subsystem communication was challenging. Various aspects of harness length and routing were not communicated effectively between the structures subsystem and avionics subsystem, leading to harnesses lengths that were in one instance too long and in another too short. Other communication challenges resulted in increased design time due to lack of explicit connector labeling and the IMUs becoming disconnected in vibe due to miscommunication about the delivered state of the hardware. Additional

emphasis will be placed on defining these interfaces and clearly communicating them, including simplified system-level wiring diagrams.

Testing and Integration

Several issues weren't apparent until integrated testing. During integration, several components were damaged or destroyed. In a typical project, this could lead to a significant amount of delay as there are typically low stocks of components. R5's emphasis on low component cost enables it to use a "parts rich" philosophy where there are many spares of all components. This allows the project to recover from all of the incidents in a manner of hours.

While performing integrated testing with the UP Board and the rest of the FlatSat, it was noted that certain components connected to the UP Board GPIO were briefly powering up or changing state while the UP Board booted. This was traced to the UP Board's GPIO behavior on start-up, with floating voltage on pins and undeclared states that can happen briefly while booting. Basic input-output settings (BIOS) configuration needed to be changed to allow for communication and pull-down resistors needed be added to GPIO pins to prevent un-commanded behavior from peripherals using these pins.

During FlatSat testing, it was discovered that some of the COTS components' power specifications did not match the power specifications listed in their datasheets. Powering on the modem drew high inrush current with transients up to 7 A, which was above the MPQ6610 load switch overcurrent protection threshold and would disable power to the modem. This was remedied by adding in-line power resistors to slow the on-time and reduce inrush current, but future designs should include soft-start load switch circuitry to control the voltage ramp rate. This highlighted the need to characterize power draw of every load in the system as measured performance may differ from specification sheets. In the future, every significant load on the power system will have its voltage and current profiles recorded through startup and any operational states, and necessary power controls implemented including soft-start or current limiting circuitry.

During testing, the inability to access digital busses outside the spacecraft made troubleshooting avionics very challenging. Future R5 spacecrafts will make these busses available via ground support equipment that can be used after the spacecraft is fully assembled.

7. AVIONICS CHANGES AND FORWARD WORK

The next two spacecrafts in development are R5 Spacecraft 3 (R5-S3) and R5 Spacecraft 5 (R5-S5). These are currently planned to be launched together on an ELaNa mission. The avionics subsystem has incorporated the lessons learned from R5-S2 and R5-S4 and will add enhancements as described below.

Reducing Design Work

When defining the interface requirements for R5-S3 and R5-S5, it was evident that there are common communication protocols and power rails across a variety of payloads and use cases. To reduce the number of avionics redesign for each iteration of spacecraft, the functionality of the original prop board was split into different boards – a propulsion and reaction wheel board and a payload interfaces board. The PDU and propulsion and reaction wheel board are intended to be reused from spacecraft to spacecraft. The interface board would be the only PCB that would need potential modifications to interface to external customers' payloads, but by breaking out GPIO, digital busses, and load switches in all available voltages it can flexibly adapt to new peripherals and payloads by primarily altering harnesses.

Improved Packaging

With the reorganization of functionality, the avionics subsystem investigated the PC/104 specification but was not satisfied with the required board area. The custom R5 PCBs could be designed in a smaller form factor than the PC/104 size of 90x96 mm [12]. The custom R5 PCBs also enforce stacking connection as optional, providing more flexibility in physical positioning within the spacecraft, as well as adaptability for different flight computers that do not conform to the PC/104 standard. R5 also considered matching the custom base to the footprint of the UP Board but ultimately decided against it due to the potential of the flight computer changing in the future.

After much debate, R5 decided that the "base" of the avionics stack will be a 60x60 mm board with four mounting holes 5 mm from the edges. If a particular board needed to be extended, it could grow in the horizontal direction only. All three of the boards would have the option of connecting using a stacking connector or Harwin Gecko G125 connectors. The stacking connector family chosen was the Samtec QTH/QSH connectors due to the size, pin power rating, and alignment pins. This stacking connector family also comes in various heights including 5 mm, 8 mm, 11 mm, 14 mm, 16 mm, 19 mm, and 25 mm. The overall height for all three boards in the stacked configuration with 8 mm and 11 mm standoffs is 41 mm. If the PC/104 specification was used, 0.6 inch (15.24 mm) standoffs would have been required, therefore increasing the overall height to 52 mm [12]. The G125 series footprints were modified by making the mounting holes slots instead of circles. While these slots slightly increase PCB manufacturing costs, this footprint modification allows for the right angle or vertical versions of the connectors to be interchanged. By adding this flexibility, the structures subsystem can specify which connector works best for wire harness routing, bend radii, and interference of other components for each flight. The stacking connector and G125 connector are located on opposite sides of the board. This design allows the boards to be mounted with one edge against

a wall, while still providing access to up to three sides of connectors, as shown in Figure 12 and Figure 13.

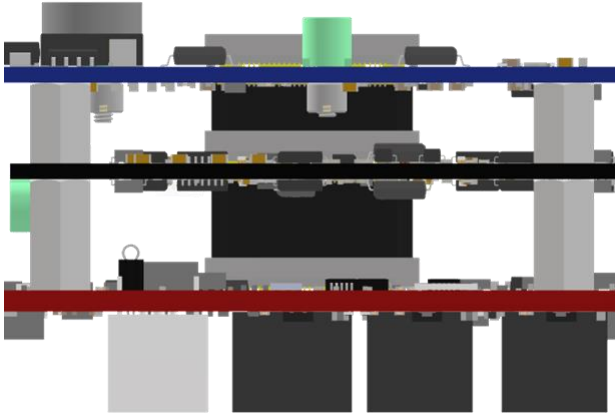


Figure 12: Next-generation avionics stack side view (top to bottom: interface board, prop/reaction wheels board, PDU)

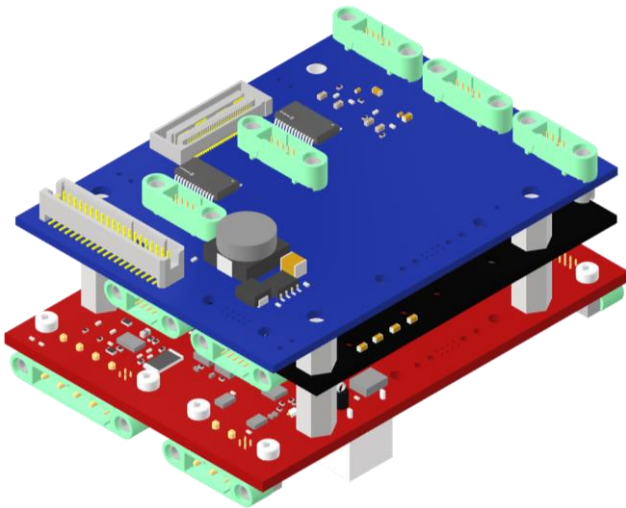


Figure 13: Next-generation avionics stack isometric view (top to bottom: interface board, prop/reaction wheels board, PDU)

Power Distribution Unit Board

Based on the challenges configuring the BQ25798, the next version of the PDU has a much simpler battery charger (TI BQ25300). The battery gauge (BQ27Z746) with the DSBGA footprint was substituted for a similar part. There are other minor changes to increase efficiency and add functionality. The Harwin M80 connectors were replaced with Harwin Geckos which simplified the wire harness process and installation tools required. The revised footprint of the PDU, Figure 14, had a surface area of 5100 mm² which is about 58% of the surface area of a PC/104 board (8825 mm²) shown in Figure 15. The PDU is also the largest of the custom R5 stack as seen in Figure 13.

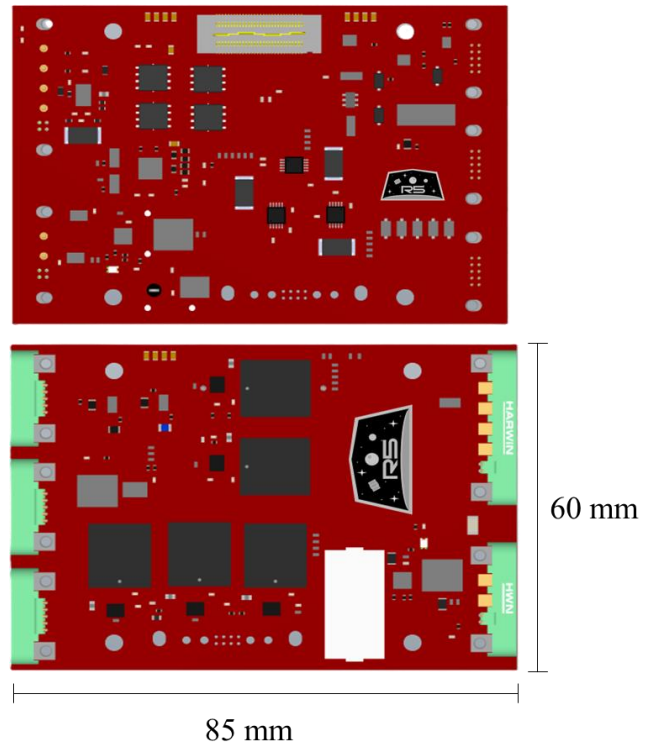


Figure 14: Top and bottom views of the PDU PCB for R5-S3 and R5-S5

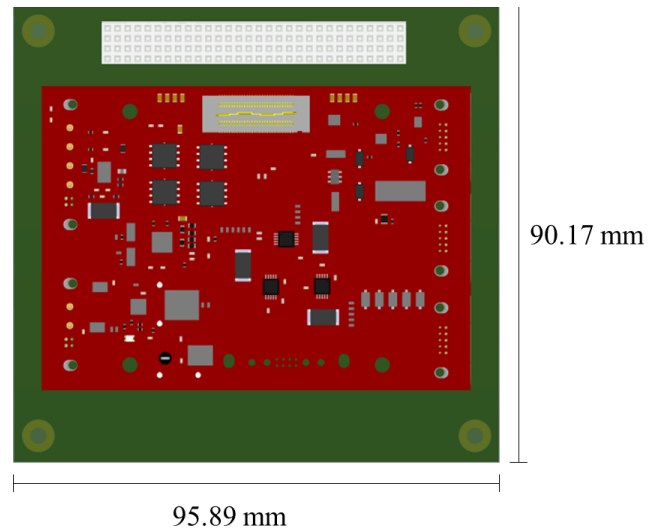


Figure 15: R5 PDU footprint (red PCB) compared to the PC/104 standard footprint (green PCB)

Propulsion and Reaction Wheels Board

On the new propulsion and reaction wheel custom PCB, there is propulsion control circuitry, as well as reaction wheel control circuitry. Learning from the previous design, the solenoid driver circuitry changed to incorporate hit and hold control of the valves. Using the TI DRV104 high-side PWM power switching driver with a 50% duty-cycle output, the power consumed when activating the thrusters or isolation

valve dropped to 25% of the power used compared to the previous design that used the MPQ6610 driver. Figure 16 shows the current draw profile of the solenoid thruster valve when controlled with the DRV104.

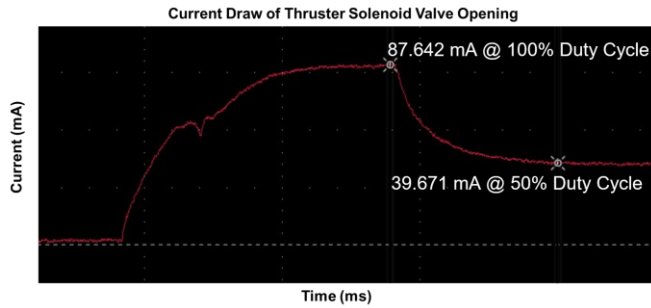


Figure 16: Current draw of a thruster solenoid valve using hit and hold driving

Additionally, a TI INA3221 power monitor is used to monitor the current draw of each thruster, providing additional data that can be used for anomaly detection and troubleshooting from the ground once in orbit. Another change from the previous design included adding light-emitting diodes (LEDs) for indicators of thrusters firing during ground testing. While these LEDs are useful in initial checkouts of the board, they do not accurately represent the current draw of a solenoid valve. These LEDs are removed for flight. A custom thruster and isolation valve load simulator board can be connected to the propulsion board instead of connecting the actual solenoid valve hardware during FlatSat testing. This will allow for LED indicators that indicate whether each valve is opened or closed and will also allow for a representative power draw of the propulsion system.

One of the main differences between S3 and S5 spacecrafts is that S3 does not contain a propulsion system. In order to keep the avionics design between spacecrafts simplified, this board will also be used for S3's reaction wheel controller. The board will not have any propulsion related components populated but will still allow for reaction wheel control functionality. The overall surface area of this board is 4191 mm², as shown in Figure 17, which is roughly 47% of the size of the PC/104 specification size.

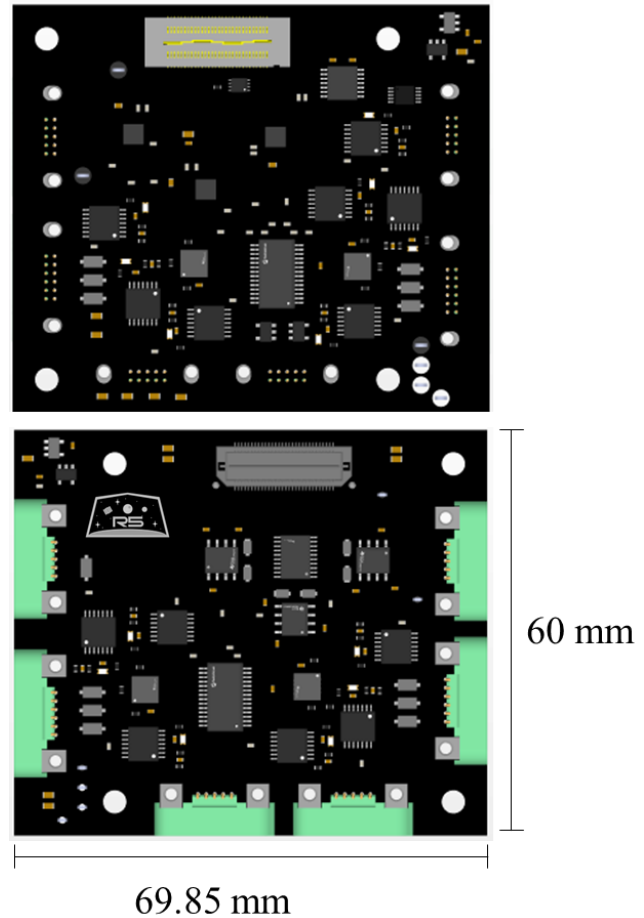


Figure 17: Top and bottom view of the propulsion/reaction wheel PCB designed for R5-S3 and R5-S5

One of R5's long-term objectives is to develop and demonstrate the foundational technologies required for in-space inspection of other vehicles and spacecraft, like proximity operations. The cold-gas propulsion system is able to provide the six degree of freedom control that's required for proximity operations, but reaction wheels are desired to minimize propellant required for attitude maneuvers. Commercially available reaction wheels intended for CubeSats are many thousands of dollars and many have long leads times, making them poor fits for R5's aggressive cost and schedule goals. For R5-S3 and R5-S5, a custom reaction wheel system will be developed. R5 is using 2.5" hard disk drives (HDDs) as they are well-balanced, high-speed, and low cost and lead time. A custom control circuit is required to drive the HDDs forward and backward at various speeds. This custom reaction wheel controller uses the AD5254 digital potentiometer to set the duty cycle of a PWM signal generated by a TLC555 timer. This PWM signal is the input to the DRV10963 three-phase brushless DC (BLDC) motor driver. One of the challenges with using HDDs is that they are sensorless and produce an extremely low back-emf when powered. If the DRV10963 cannot detect the back-emf, the initial start-up commutation sequence is not accurate for very low speeds.

In order to control and enable the reaction wheels and propulsion components, GPIO signals are used. Since the UP Board has limited GPIO signals available, two additional MCP23017 I2C GPIO expanders were added. One GPIO expander is used for reaction wheel control while the other is used for control of thrusters and the isolation valve in the propulsion system.

The last major change from the previous design to the new design of propulsion control is the use of the MCP3221 12-bit analog to digital converter (ADC) that is used for the pressure transducer. In the previous design, the UP Board GPIO ADC functionality was used with the pressure transducer. This ADC was an 8-bit reading from 0 to 3.3 V. With an 8-bit ADC, the resolution is 19.5 mV/bit, which equates to 24.375 psi/bit. The 12-bit ADC increases that resolution to 1.22 mV/bit, which equates to 1.525 psi/bit. This increased resolution will provide greater performance data of the propulsion system while in orbit.

Interface Board

A separate PCB will be used as the interface board. This board interfaces the UP Board and PDU with the R5 core avionics as well as external payloads to provide power and signal.

The same I2C-controlled MCP23017 GPIO expanders found on the prop board are used to provide a bank of pins for R5 avionics functionality and a bank for external payloads to utilize. Broken out to dedicated payload connectors in the center of the interface board, these payload GPIO can be flexibly allocated by wire harness design changes to support various payloads. On R5-S5, the interface board payload GPIO bank will provide several output signals to a hosted payload to enable its different operations, as well as a UART interface for additional data and commanding.

With propulsion-specific controls and power moved to the prop board, the interface board will be used to switch power to the remaining R5 components and payloads. MAX20313 programmable-current load switches are being used for 3.3 V and 5 V loads, namely the cameras, modem, GPS, and IMU. These replace the MPQ6610 half-bridge drivers due to the more flexible 0.5 A to 6 A adjustable current limit and continuous current-limit mode during an overcurrent event. Enable signals to the MAX20313s are provided from the R5 avionics GPIO bank, which guarantees known output states during computer bootup and a “fail-off” mode for switches.

12 V power is also being provided to the hosted payload, so an MPQ5872 is used as a load switch, with its higher voltage limit of 28 V. The payload contains a nichrome burn wire, which heats up when 5V is applied to melt a monofilament line. This required a 5V line capable of limiting its output current when presented with a “short” that the nichrome appears as. To prevent a load switch from going to overcurrent protection mode, an ADP2303 step-down DC-DC regulator will be used in 5 V, 3 A output mode. When enabled, it will provide a constant 3 A to the nichrome wire

to heat it rapidly but will not trip overcurrent protection and shut off as most switches would.

The overall surface area of the interface board is 4800 mm², as shown in Figure 18, which is roughly 54% of the size of the PC/104 specification size. It has gecko connectors to break out power and signal to R5 radiation detection boards, modem and cameras, GPS and IMU, and payloads.

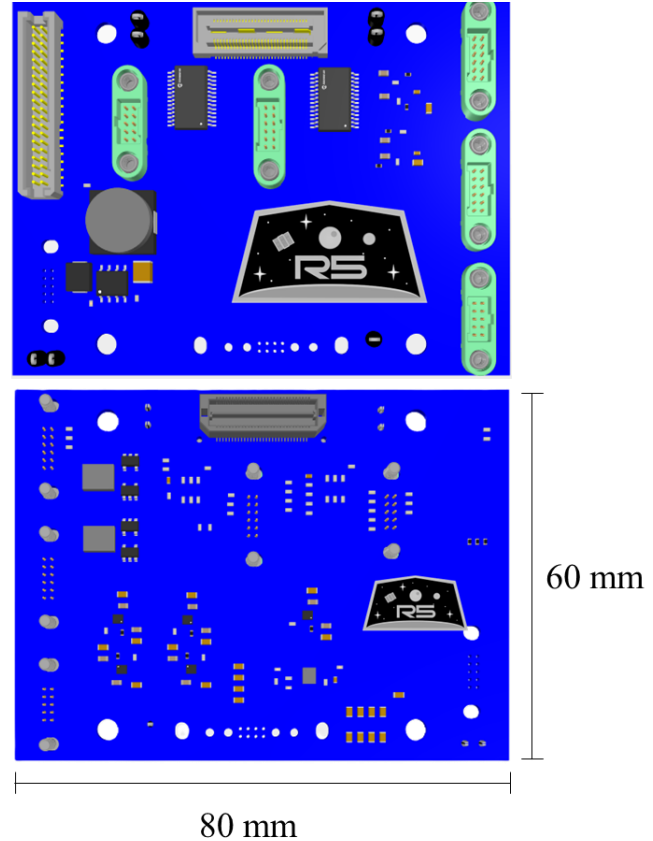


Figure 18: Top and bottom views of the interface board PCB designed for R5-S3 and R5-S5

8. SUMMARY

R5 has successfully rapidly and affordably designed, built, tested, delivered, launched, and operated two spacecrafts. This suggests that the lean process it is pioneering may be suitable to enable risk-accepting programs to substantially reduce their cost and schedule for getting nascent technologies demonstrated in orbit. R5 was able to do this with an iterative and incremental development paradigm creating many lessons learned along the way. The avionics subsystem identified functionality of the spacecrafts that couldn't be implemented within the other constraints of the project and subsequently iteratively developed a process that resulted in the successful operation of custom components in orbit for six months and counting. R5 will continue to gather more data about the suitability of this process with the continued operation of R5-S2 and R5-S4 and additional

missions currently in active development. The project will continue to share its findings with the community with the goal of enabling others to enhance their spacecraft capabilities while also reducing their cost and schedule. Several spacecraft launches are planned over the next couple of years, which will demonstrate the avionics advances described above in addition to refinements of the R5 process.

REFERENCES

- [1] *CubeSat Design Specification*, Version 14.1, 2022. [Online]. Available: https://static1.squarespace.com/static/5418c831e4b0fa4ecac1bacd/t/62193b7fc9e72e0053f00910/1645820809779/CDS+REV14_1+2022-02-09.pdf
- [2] Sam Pedrotty, “Single Board Computer Radiation Test Results and Radiation Test Software,” presented at the 43rd AAS GNC Conference, Breckenridge, CO, USA, February, 2020.
- [3] Sam Pedrotty, Jake Sullivan, Elisabeth Gambone, and Thomas Kirven, “Seeker 1.0 GNC Flight Performance,” presented at the 43rd AAS GNC Conference, Breckenridge, CO, USA, February, 2020.
- [4] Jake Sullivan, Elisabeth Gambone, Thomas Kirven, and Sam Pedrotty, “RAPID DEVELOPMENT OF THE SEEKER FREE-FLYING INSPECTOR GUIDANCE, NAVIGATION, AND CONTROL SYSTEM,” presented at the 42nd AAS GNC Conference, Breckenridge, CO, USA, February, 2019.
- [5] Jay N. Estes, Sam Pedrotty, and Ronald R. Sostaric, “SPICE Flight Hardware Environmental Test Qualification Program,” in AIAA Scitech 2021 Forum, 2021, doi: <https://doi.org/10.2514/6.2021-0257>.
- [6] NASA, J27301 Materials and Processes Selection, Control, and Implementation Plan for JSC Flight Hardware, Rev F, Johnson Space Center, Houston, TX, Aug. 2009. [Online]. Available: <https://standards.nasa.gov/sites/default/files/standards/JSC/F/0/JSC-27301F.pdf>.
- [7] IPC-6012-ES, *Qualification and Performance Specification for Rigid Printed Boards*, Rev. E, Addendum S, Institute for Printed Circuits, Apr. 1, 2020, pp. 1-20.
- [8] Voltaic Systems, “P123” [Online]. Available: https://voltaicsystems.com/content/P123_ds.pdf
- [9] Voltaic Systems, “P124” [Online]. Available: https://voltaicsystems.com/content/P124_ds.pdf
- [10] *EEE Parts Management and Control for MSFC Space Flight Hardware*, MSFC-STD-3012
- [11] *Electrical Grounding Architecture for Unmanned Spacecraft*, NASA-HDBK-4001
- [12] *PC/104 Specification*, Version 2.6, PC/104 Embedded Consortium, 2008. [Online]. Available: https://pc104.org/wp-content/uploads/2015/02/PC104_Spec_v2_6.pdf

BIOGRAPHY



Kathryn Knesek received a B.S. in Electrical Engineering from Virginia Tech in 2022 and is currently pursuing an M.S. in Space Systems Engineering at The Johns Hopkins University. She has been with NASA for five years as an electronics engineer, supporting the xEMU, R5, Orion DFI, and Gateway WAP projects.



Morgan Alexander received a B.S. in Electrical Engineering from the University of North Texas in 2020. She has been with NASA for six years as an electronics design engineer supporting various projects.



Jack Wisbiski received a B.S. in Electrical Engineering in 2020 and a M.S. in Electrical and Computer Engineering in 2021 from the University of Michigan. He has been with NASA for five years as an electronics engineer, doing design and test work in support of R5, SPLICE, MPG, and Orion DFI projects.